

Enhancing McAdams Coefficient-Based Speaker Anonymisation with Cross-Gender Timbre Transfer

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Abstract—Speaker anonymisation or speaker de-identification involves modifying original speech to resemble the voice of an unspecified speaker, while preserving linguistic content and speech quality. This study introduces a speaker anonymisation system based on the second baseline system from the VoicePrivacy2022 Challenge. The evaluation was performed using the evaluation scripts provided by the Challenge. Enhanced privacy is achieved by using a VAE-GAN timbre transfer model to disguise gender identity through a random gender selection strategy. Additionally, the primary objective utility evaluation shows the potential for further improvements. In the secondary utility evaluation, the proposed system shows favourable results with respect to voice distinctiveness, surpassing all baseline systems.

Index terms—Speaker anonymisation, Pseudonymisation, Privacy, Anonymity, Automatic speech recognition, Automatic speaker verification

I. INTRODUCTION

Recently, there has been increased attention on the potential privacy risks in the evolving digital landscape due to the quick advancements of global network systems and neural networks. Privacy risks encompass a number of dimensions, with surveillance and eavesdropping risks being two of the most prominent in the context of the development of various technologies under artificial intelligence technologies, which often manifest themselves in the excessive collection and misuse of user data.

Speaker anonymisation or speaker de-identification is a method of altering original speech in order to resemble the voice of an anonymous speaker, while retaining the linguistic content and quality of speech [11]. In addition, as a form of biological information, human speech contains a wide range of personal information that can be revealed by automated systems [22]. This information encompasses details such as the individual's age, gender, racial or ethnic background, place of origin, health or emotional condition, political affiliations, and other delicate information [9][33]. Furthermore, speaker anonymisation differs from speech anonymisation as it aims to provide a more profound level of anonymisation than the latter. Speaker anonymisation conceals the speaker's identity, while speech anonymisation only conceals the linguistic content [11]. For all participants in the VoicePrivacy Challenge 2022, speaker-level anonymisation is a fundamental evaluation criterion. This means that all utterances of each speaker must be transformed into the corresponding pseudo-speaker [33].

It is now proposed that building a speech synthesis system from a vast number of speaker utterances is a straightforward process. Such a system can perform sound cloning, generating new samples for a particular speaker that are more responsive to their speech characteristics [16][11]. Moreover, an automatic speaker identification system can unveil the speaker's identity, which is the most significant aspect [32]. For instance, large technology companies and even government agencies have consistently conducted surveillance activities to analyse user identities and collect desired information [5][12].

In line with the VoicePrivacy2022 initiative and challenge, this work aims to further explore the potential of established signal processing techniques and neural-style timbre transfer model as a solution to the speaker anonymisation task by disguising gender identity. By doing so, we try to illuminate the evolving privacy concerns and propose effective solutions to defend the individual data and privacy.

The paper has the following structure: Section 2 discusses previous research on speaker anonymisation, including the x-vector based methods, voice-conversion based methods, and the generalised digital signal processing approach. The proposed anonymisation system is presented in Section 3. In Section 4, the dataset's composition, privacy metrics most relevant for VoicePrivacy Challenge 2022, along with the results of the experiments and investigation, are described. The overall experimental results and main conclusions are discussed in Section 5. In Section 6, possible scenarios for further improvement of this research are discussed.

II. PREVIOUS WORK

Speaker anonymisation can be classified into physical or logical anonymisation, depending on the intended manipulation goals [11]. Physical anonymisation includes introducing an external sound to the original waveform or applying voice-conversion-based digital signal processing techniques to alter some part of the speech, such as pitch, loudness, or timbral characteristics. Logical anonymisation involves modifying the identity of the speaker while also keeping the identity of the pseudonymised person readily available at all times. Alternatively, digital signal processing techniques can also implement logical anonymisation, as the second baseline does, to achieve the results of speaker anonymisation [26].

A. X-vector Based Methods

The B1 system was developed using the state-of-the-art x-vector method and a waveform model. The concept behind this anonymised system is to extract distinctive x-vectors from speech, isolate them from the phoneme posterior map and F0 information, and then modify these x-vectors. Ultimately, an acoustic model will produce the final anonymised speech by integrating the previously obtained F0 and phoneme posterior data using a neural source filter model.

B. Voice-Conversion Based Methods

Jin et al. [14] proposed a system based on voice conversion or transformation. This system aims to conceal the identity of a speaker when communicating with another two speakers. Likewise, Bahmaninezhad et al. [3] introduced a de-identification system that uses a convolutional neural network (CNN) encoder-decoder architecture. This system maps the voice of a specific speaker to the average voice of all speakers in a speaker pool (or the average voice based on gender).

Alegre et al. [2] investigated how voice conversion techniques can help to evade surveillance biometric automatic speaker verification systems. They assessed the effect of obfuscation on various classic ASV systems, with results showing that the standard GMM-UBM system is the most susceptible, exhibiting a decline from 9% EER to 34% EER. Meanwhile, the IV-PLDA system is the most robust system. The voice conversion techniques they employed was the Gaussian dependent filtering (GDF) approach.

Vocal tract length normalization (VTLN) techniques [8] are applied to the anonymisation task by Hidebehind [27]. A warping function, when applied to the spectral logarithmic amplitude of each speech frame, can alter both the spectral envelope and pitch of the processed speech. Patino et al. [26] noted that this method provides stronger anonymisation, and in case of human listener identification, the B2 system (McAdams coefficient-based system) may be inferior. Moreover, they suggest that pitch changes are unnecessary if an anonymisation system is designed solely to deceive the automatic speaker verification.

Additionally, the T32 system proposed by Mawalim et al. [17] is based on Time-Scale Modifications (TSMs) for pitch shifting. Mawalim et al. [18] also presented a new system, based on B2, to improve the security of the speaker anonymisation system. This is achieved by using speech watermarking. This method is comprised of two phases, embedding and detection. In the embedding phase, the two different McAdams coefficients are represented by 0 and 1. The watermarked speech is then obtained by bit inverse switching, frame-by-frame. In the detection phase, a comparison of the power spectrum is performed.

System T04 proposed by Meyer et al. [20] achieves speaker anonymity through three steps: converting audio to phonetic transcriptions, creating new and distinct voices using a Generative Adversarial Network (GAN), and synthesizing anonymous speech based on transcriptions and speaker embeddings.

C. Neural Speech Style Conversion Methods

Speech style conversion refers to the process of synthesising speech samples from one source speaker to another target speaker, where the linguistic content and speech style should be kept constant [1].

AlBadawy and Lyu [1] proposed a voice-conversion model based on the VAE-GAN architecture in the time-frequency domain. Their approach is an extension of the UNIT model, where a shared latent space is facilitated by a universal encoder [15][4]. Their approach involves the use of a unified encoder across all source speakers, which enhances the capacity for effective generalisation to multiple target speakers. They also incorporate a latent loss mechanism to constrain the encoded features, effectively obfuscating the identity of the source speaker. The result is remarkably natural synthesised speech that seamlessly captures the distinctive style of each speaker.

D. Systems Considering the Gender Factor

In general, a speech anonymisation system dealing with the gender factor will use three strategies: same, opposite and random. Srivastava et al. [29] performed an empirical investigation of same and opposite gender selection.

Participant T09 proposed an anonymisation strategy for X-vectors based on gender selection: same, opposite, random, which ranks first in terms of subjective privacy evaluation.

Stoidis and Cavallaro [30] present a new method for encoding gender information and disentangles gender and identity information by creating an embedding space z that independently captures gender information. Speech can be reconstructed to match the identity and/or gender of the intended speaker.

Another speech anonymisation system developed by Stoidis and Cavallaro [31] utilises a GenGAN architecture. The proposed network comprises a U-Net generator $\mathcal{G}$ architecture which learns to deceive the discriminator $\mathcal{D}$. The generator $\mathcal{G}$ is conditioned solely on gender information, and they employ an adversarial generator loss function $\mathcal{L}_{\mathcal{G}}$ to achieve a balance between signal distortion and privacy preservation by learning a non-binary gender. The generator loss is calculated as:

$$\mathcal{L}_{\mathcal{G}} = \mathcal{L}_d(\mathcal{M}, \mathcal{M}') + \epsilon \mathcal{L}_a(\mathcal{Y}, \mathcal{Y}_F) \quad (1)$$

where ϵ is the dis-utility budget [34] in the privacy-utility trade-off.

Noé et al. [24] introduced a system that relies on speaker embedding, whereby gender information is concealed by transforming the embedding and pitch.

Champion et al. [6] presented a new method in their study to improve the privacy of a speech pseudonymization system based on x-vectors. This was achieved by modifying the F0 paralinguistic information. The results of the experiments suggest that modifying the F0 and choosing an x-vector from the *opposite* gender offer improved privacy protection against a

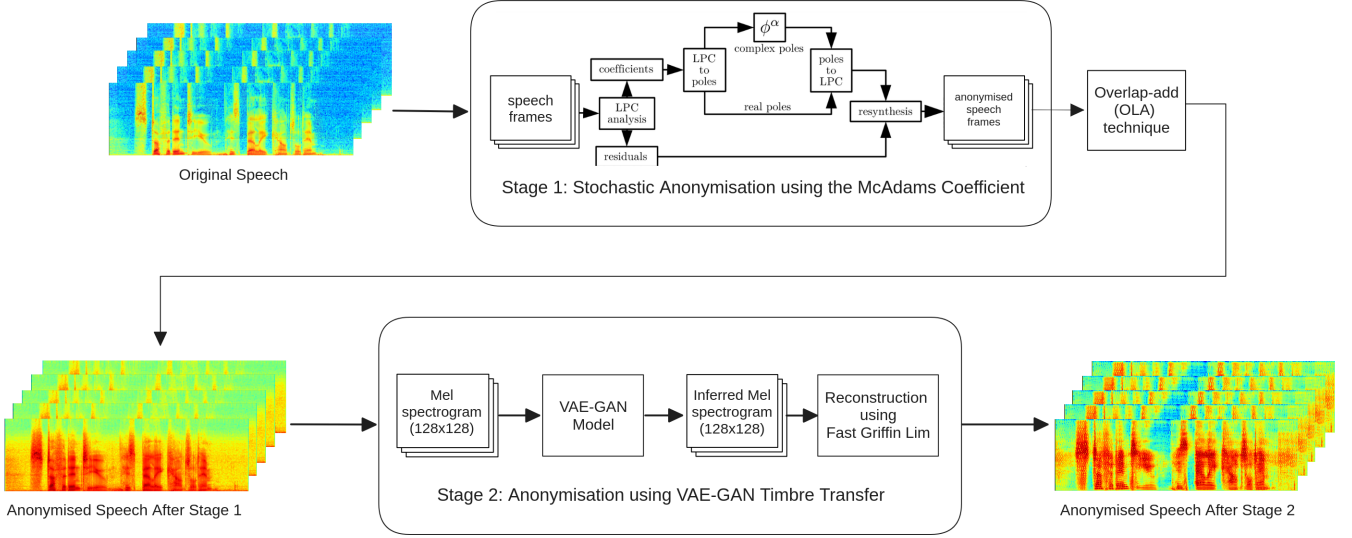


Figure 1. The overall architecture of the proposed system. The original speech is used as input and frame-by-frame computation is performed through the Stage 1 to output the result of the first anonymisation. In the Stage 2, the 128x128 mel-spectrograms of the anonymised speech after Stage 1 are extracted. Then the newly generated 128x128 mel-spectrograms are inferred by the pre-trained VAE-GAN model. Finally, the signal is reconstructed using the Fast Griffin-Lim algorithm. The example waveform in the figure is 1089-134686-0001.wav from *libri_test_trials_m*. Some parts of the diagram of the Stage 1 is taken from [26].

semi-informed attacker. They also found that performance of the anonymisation system depends on the gender of the source.

III. SYSTEM OVERVIEW

The main motivation for developing this anonymisation system was to explore the possibilities of disguising gender information using a pre-trained VAE-GAN timbre transfer model. This approach was combined with the B2 system in order to merge together as the full structure of this work [26].

Furthermore, this work was developed using digital signal processing methods rather than the x-vector methods. The VAE-GAN timbre transfer model also uses pre-trained models to optimise cost-effectiveness, especially in the area of speaker anonymisation. This method does not require any training data in the anonymisation phase, providing a relatively efficient and lightweight alternative compared to x-vector based methods. However, not using pre-trained models and opting for self-training would result in additional training costs. In addition, the entire evaluation process requires training of the ASV and ASR systems.

A. Architecture

The overall architecture of the proposed system is shown in Figure 1., which consists of two stages: the McAdams coefficient-based anonymisation method and the VAE-GAN timbre transfer model. Figure 6. shows the spectrogram and mel-spectrogram of a given original utterance, together with its corresponding speech sample, after being anonymised by stage 1 and then by stage 2.

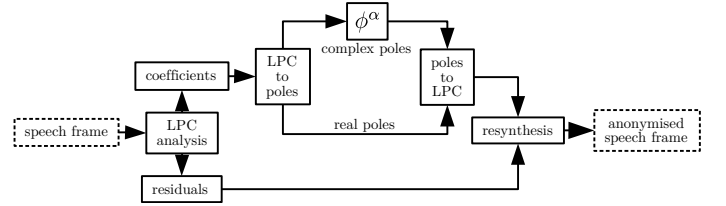


Figure 2. The architecture of the McAdams coefficient-based anonymisation system. [26]

B. Stochastic Anonymisation using the McAdams coefficient-based Method

The McAdams coefficient has a strong relationship with the timbral characteristics due to its association with harmonic frequency distributions [19]. This is derived from a parameter in an additive synthesis formula used in music signal processing [10]. Multiple harmonic cosine oscillations can be reconstructed using this method to generate timbre.

$$y_{\text{syn}}(t) = \sum_{h=1}^H r_h(t) \cos(2\pi(h f_0)^\alpha t + \Phi_h) \quad (2)$$

where α represents the McAdams coefficient, h is the harmonic index, $r_h(t)$ is the amplitude and Φ_h is the phase [19]. The synthesized signal $y_{\text{syn}}(t)$ is produced by combining multiple harmonic consinusoidal oscillations with different frequencies and amplitudes. This is carried out through the inverse Fourier series representation.

As proposed in the paper of the B2 system, adjusting the McAdams coefficient has a significant impact on the spectral envelope of the speech signal, thus affecting how we perceive the timbre of the sound. Non-harmonic components, also referred to as overtones or partials, arise from applying harmonics to additive synthesis.

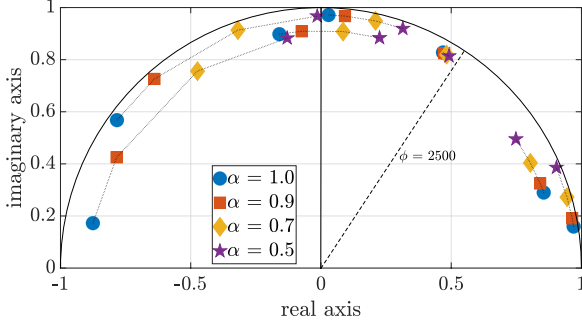


Figure 3. Pole positions and the effect of pole shifting in the z plane, where $\alpha \in \{0.9, 0.7, 0.5\}$. [26]

The basic idea behind this method is to use McAdams coefficients to adjust the formant position by adjusting the pole positions obtained when analysing the speech signal using Linear Predictive Coding (LPC). Altering the formant position in this manner could potentially result in a decrease in the performance of the ASV system when speech is anonymised.

The general process of the McAdams coefficient-based anonymisation system is presented in Figure 2. [26]. As demonstrated, the process begins by employing a frame-by-frame LPC source-filter analysis to derive the LPC coefficients and residuals. Polynomial root-finding is employed on the LPC coefficients to produce pole positions in the z -plane. These poles include complex poles with nonzero imaginary terms and real poles with zero imaginary terms. Also, every pole matches a peak in the spectrum, expressed in human speech as a cluster of the formant position. The residues are being prepared for the final re-synthesis in the time domain.

The McAdams transform is applied to the phase of each pole, while preserving the imaginary poles. The phase ϕ (between 0 and π radians) of the poles with non-zero imaginary parts is raised to the power of the McAdams coefficient. As a result, shifting the position ϕ^α of these complex poles causes the angle to shift either clockwise or counterclockwise of the complex positions.

Figure 3. [26] shows an example of the original pole position, as well as the shifted pole positions when $\alpha < 1$ for $\alpha \in \{0.9, 0.7, 0.5\}$. The scale and the direction of the shift is determined by both angle and the McAdams coefficient α .

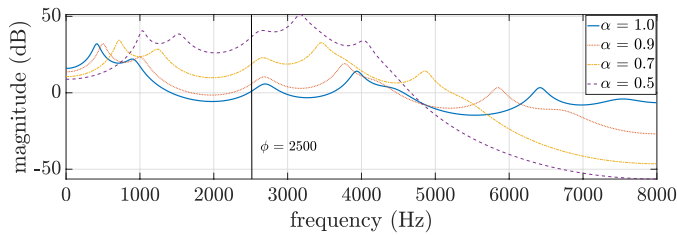


Figure 4. Spectral envelopes of an utterance frame and pole positions for McAdams coefficient $\alpha \in \{0.9, 0.7, 0.5\}$. The value of $\phi = 1$ radian corresponds to the frequency of 2500 Hz. [26]

The effect of the angle shifting in the spectral envelope is shown in Figure 4. The blue lines represent the original speech spectrum, while the yellow and purple lines represent the anonymised speech spectrum after applying the anonymisation process by the corresponding McAdams coefficient.

C. Anonymisation using VAE-GAN Timbre Transfer Model

iii. Pre-Trained VAE-GAN Model

This model is an extension of the VAE-GAN model from AlBadawy and Lyu [1], which is itself an extension of the unsupervised image-to-image translation networks, UNIT [15]. This study employs the pre-trained VAE-GAN timbre transfer model [4] by executing the inference procedure then output the final anonymised speech audio.

Figure 5. shows the design of a single path in the VAE-GAN timbre transfer model. This model includes a universal encoder, whose input is the real spectrogram, and a shared residual block for decoding at a shallow level, as well as several decoder-discriminator pairs specific to target domains.

Each decoder-discriminator pair corresponds to a particular sound characteristic or style, allowing the blending of the abstract representation extracted by the universal encoder with the information specific to these target domains, resulting in transformation outputs with distinct sound characteristics.

The VAE-GAN timbre transfer model was pre-trained at Flickr 8k Audio Caption Corpus [13], which provided by [4]. In this work, the two paths refer to the transformation from the voice of a female speaker to that of a male speaker, as well as the transformation from the voice of a male speaker to that of a female speaker.

iii. Gender Selection Strategy: Random

In general, a speech anonymisation system that considers gender makes use of three strategies: same, opposite and random. An experimental study of same and opposite selection was conducted by Srivastava et al. [29].

Similar to the stochastic anonymisation approach used in the B2 system, the *target gender identity* is randomly generated based on the random seed when anonymising the speaker for the stage 2. Each speaker will all correspond to a unique *target gender identity* and McAdams coefficient that is randomly generated. The *source gender identity* is determined by the actual gender of the speaker as specified in the *spk2gender* file.

Furthermore, the automatic speaker verification system that the attacker intends to retrain at the *utterance-level* still determines the *source gender identity* based on the actual gender of the *speaker of interest*. This assumption is based on the belief that the attacker can accurately determine the actual gender of the *speaker of interest* for each utterance, which is expected to hold true in most cases. In addition, the *target gender identity* is completely randomised at the *utterance-level*.

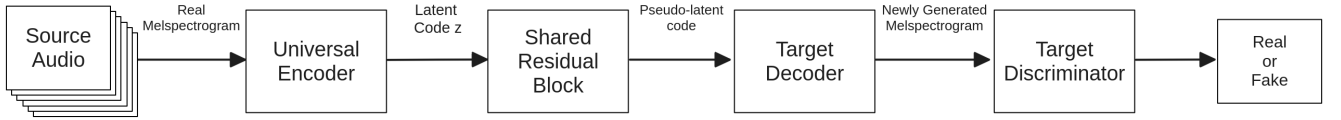


Figure 5. The design of the VAE-GAN timbre transfer model. The aim of the universal encoder is to extract the feature of the source speaker rather than the timbre, while the decoder aims to introduce the timbre of the target speaker into the content as motivated by the discriminator. This diagram is taken from [4].

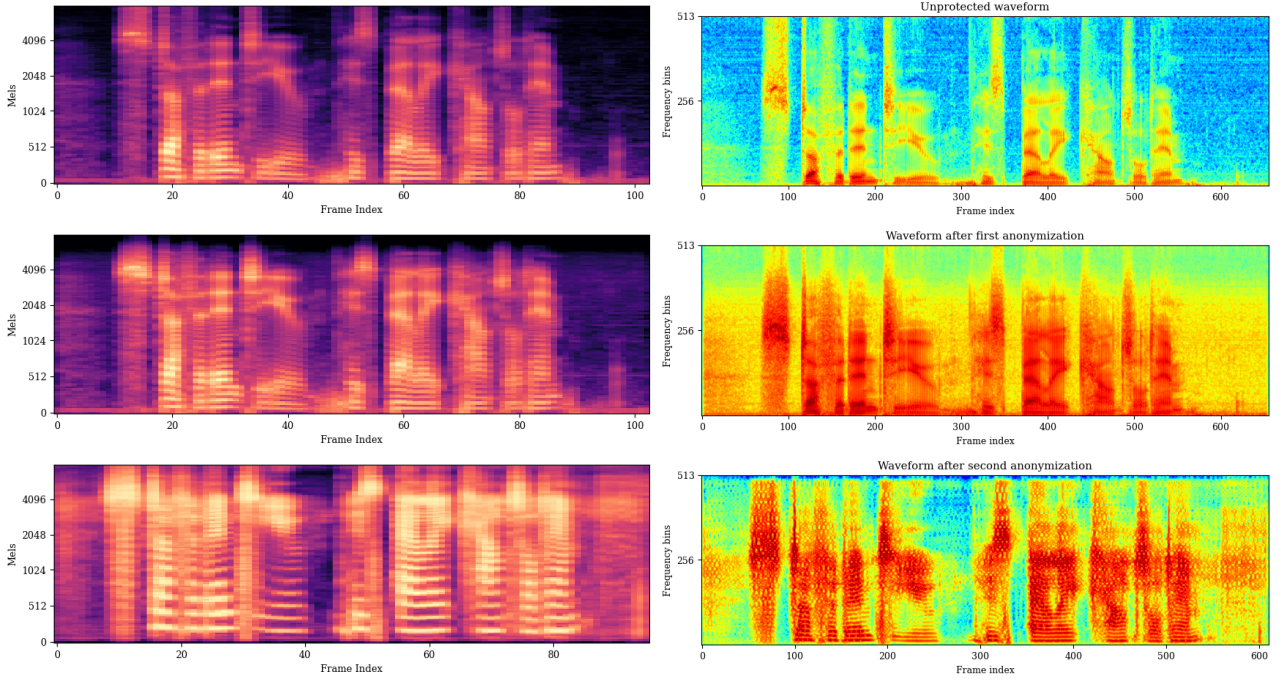


Figure 6. These are the mel-spectrograms and spectrograms of three speech samples: the original speech from 1089-134686-0001.wav (from *libri_test_trials_m*), the speech after anonymisation using the McAdams Coefficient method, and the speech after timbre transfer.

In this paper, we focus on the impact of *random* gender selection on evaluation in a voice conversion anonymisation system. Figure 6. illustrates the mel-spectrogram and the spectrogram of the utterance after two stages. The example used is utterance sample 1089-134686-0001.wav from the *LibriSpeech* test dataset, spoken by a male speaker, with the *target gender identity* set to female.

IV. EXPERIMENTS SETTINGS AND RESULTS

Unlike the original setting of B2, in this experiment the McAdams coefficient α is randomly chosen from the range 0.65 to 0.85.

A. Attack Models & Scenarios

Within the attack model's context, the adversary has access to diverse speech samples spoken by a particular individual, which we will call the *speaker of interest*. These speech samples serve as reference points and are termed as *enrollment utter-*

ances. These samples could be in their original form or may have undergone anonymisation.

In general, an attacker of a privacy system can be categorised as an *ignorant* attacker, a *semi-informed* attacker, and an *informed* attacker. Here the attacker is assumed as *semi-informed*, i.e., this attacker will not be as knowledgeable about the algorithm and configuration as a *informed* attacker.

For evaluation, there are three attack models:

- The attacker has access to a set of trial utterances, which have undergone anonymisation.
- The attacker have access to the original or anonymised enrollment utterances for each speaker.
- The attacker is assumed to retrain the automatic speaker verification system on anonymised data on a *utterance-level*.

B. Datasets

All datasets used in this work were specified by the VoicePrivacy 2022 Challenge. Subsets from *LibriSpeech* [25], *LibriTTS* [36], *VCTK* [35] and *VoxCeleb-1,2* [21] [7] were selected and

allocated for use at different stages. The ASR_{eval} , ASV_{eval} , ASR_{eval}^{anon} and ASV_{eval}^{anon} systems are trained on the *LibriSpeech train-clean-360* dataset.

Development Set		F	M	ALL	Utterances
LibriSpeech dev-clean	Enrollment	15	14	29	343
	Trial	20	20	40	1978
VCTK-dev	Enrollment	15	15	30	600
	Trial (different)				10677
	Trial (common)				695

Table 1. Composition of the Development set.

Evaluation Set		F	M	ALL	Utterances
LibriSpeech test-clean	Enrollment	16	13	29	438
	Trial	20	20	40	1496
VCTK-test	Enrollment	15	15	30	600
	Trial (different)				10748
	Trial (common)				700

Table 2. Composition of the Evaluation set.

The McAdams coefficient-based method does not require a training set. A detailed breakdown of the number of speakers and utterances within these two datasets is provided in Tables 1-2.

C. Privacy and Utility Metrics

Primary privacy and utility metrics include objective privacy evaluation and objective utility evaluation using ASV and ASR respectively. Secondary utility metrics include pitch correlation score and voice distinctiveness gain.

iii. Primary Objective Privacy Evaluation: EER, ASV

The primary objective privacy evaluation metric EER% (Equal Error Rate in %) is obtained by the execution of two automatic speaker verification (ASV) systems, ASV_{eval} and ASV_{eval}^{anon} . Higher EER indicates better level of privacy.

The input to the ASV_{eval} system are the original utterances. The ASV model used by the semi-informed attacker is ASV_{eval}^{anon} , which is trained on the anonymised version of *train-clean-360* from *LibriSpeech*, and is used to determine whether two anonymised utterance authors, pseudo-speaker A and pseudo-speaker B, are the same person.

The ASV_{eval}^{anon} system performs speaker verification task by extracting x-vectors speaker embeddings and computing probabilistic linear discriminant analysis (PLDA). X-vectors of each pair of enrollment and trial utterance are obtained and then output a log-likelihood ratio (LLR) score. The LLR score will be used in the form of a threshold θ to determine whether or not the two utterances were made by the same person.

The EER has the form of

$$EER = P_{fa}(\theta_{EER}) = P_{miss}(\theta_{EER}) \quad (3)$$

where P_{fa} is the false alarm at threshold θ , P_{miss} is the miss rates. The EER is computed when the false alarm P_{fa} and the miss rates P_{miss} are equal to each other at the same threshold θ_{EER} .

iii. Primary Objective Utility Evaluation: WER, ASR

Similarly, the primary objective utility evaluation is performed by two automatic speech recognition (ASV) systems, ASR_{eval} and ASR_{eval}^{anon} . The input to the ASR_{eval} system is also the original utterances, and the ASR_{eval}^{anon} system is trained on the anonymised version of *train-clean-360* from *LibriSpeech*. The ASR_{eval}^{anon} system outputs the recognised results at the end to calculate the WER% (Word Error Rate in %) for the primary objective utility evaluation.

The WER is computed as

$$WER = \frac{N_{substitution} + N_{deletion} + N_{insertion}}{N_{reference}} \quad (4)$$

where $N_{substitution}$, $N_{deletion}$ and $N_{insertion}$ represents the error number of their respective subscripts.

The performance of the ASR_{eval}^{anon} system is expressed in terms of WER, with lower WER representing better utility of the designed anonymisation system and vice versa.

iii. Secondary Utility Evaluation: p^{F_0} and G_{VD}

The pitch correlation p^{F_0} between original and anonymised utterances is used to evaluate how much the intonation of the anonymised voice changes. The pitch correlation p^{F_0} of two identical utterances is equal to 1.

The gain of voice distinctiveness G_{VD} is designed to evaluate the voice distinctiveness between the anonymised speech and the original speech:

$$G_{VD} = 10 \log_{10} \left(\frac{D_{diag}(M_{aa})}{D_{diag}(M_{oo})} \right) \quad (5)$$

where M_{aa} is the anonymised utterance matrix and M_{oo} is the original utterance matrix. The gain of voice distinctiveness G_{VD} of two identical utterances is equal to 0. A clear diagonal means distinguishable sound, so in terms of utility, better privacy means, higher G_{VD} .

D. Results Summary

This evaluation was performed based on the evaluation scripts provided by the VoicePrivacy Challenge 2022.

iv. Primary Privacy Evaluation Results

The results of the primary privacy evaluation, expressed in EER%, are shown in Table 3.

As can be seen from the Table 3, the EER% scores of this system are higher than those of the B2 system for all datasets, except for the male subset of *VCTK-dev (common)*, the female

Dataset	F/M	Weight	EER%		
			Orig	B2	Proposed
LibriSpeech-dev	F	0.25	8.67	11.36	11.93
	M	0.25	1.24	1.40	2.95
VCTK-dev (different)	F	0.20	2.86	6.68	11.29
	M	0.20	1.44	6.35	8.39
VCTK-dev (common)	F	0.05	2.62	5.81	7.27
	M	0.05	1.43	8.83	6.55
Weighted Average Dev Set			3.54	6.53	8.35
LibriSpeech-test	F	0.25	7.66	7.12	7.85
	M	0.25	1.11	1.11	2.00
VCTK-test (different)	F	0.20	4.89	16.92	13.68
	M	0.20	2.07	7.69	8.61
VCTK-test (common)	F	0.05	2.89	10.98	8.96
	M	0.05	1.13	4.80	5.93
Weighted Average Test Set			3.79	7.77	7.67

Table 3. Primary objective privacy evaluation results of metric EER% for ASV_{eval}^{anon} , compared with the original data and the second baseline.

subset of *VCTK-test (different)*, and the female subset of *VCTK-test (common)*.

Furthermore, the best performance is achieved on the *VCTK-dev (different)* dataset.

Table 4. provides a more detailed breakdown of scores for the various performance metrics used to evaluate the biometric recognition system on the *VCTK-dev (different)* dataset. Overall, it can be observed that female speakers within the same dataset tend to be more resistant to recognition by the ASV system, showing a higher EER. A more detailed description of the ZEBRA evaluation framework can be found in the ZEBRA paper [23].

iv. Primary Utility Evaluation Results

The results of the primary utility evaluation, expressed in WER%, are shown in Table 5.

It should be noted that this system is deficient in terms of WER%, i.e. the results of anonymised speech are not sufficiently intelligible and the ability of the proposed system to preserve linguistic content is limited. Comparing the performance on the *LibriSpeech* and *VCTK* datasets, we can also see that *VCTK* is a more challenging dataset, as its speakers use different English accents, pronunciations and speaking rates. This suggests that the use of a pre-trained voice transfer model may fall short in its performance when faced with the diversity of speaker speech styles.

The reason for this could be that that when the gender identity of the source and the randomly generated target are the same—either both female or both male—the speech intelligibility consistently decreases. Perhaps changing the speakers to anonymous speakers of the opposite gender could improve the utility. However, the privacy implications of such an approach remain uncertain.

Table 6. shows the various components required to calculate the WER, including the number of substitutions, deletions, insertions and the number of references.

VCTK-dev (different)	F/M	Score
EER	F	11.29%
	M	8.387%
Cllr (min/act)	F	0.377/2.242
	M	0.303/1.226
ROCCH-EER	F	11.255%
	M	8.315%
linkability	F	0.727125
	M	0.786818
Population (ZEBRA)	F	0.440 bit
	M	0.492 bit
Individual (ZEBRA)	F	3.446 (C)
	M	2.773 (C)

Table 4. Results of Cllr (min/act), ROCCH-EER, linkability, population (ZEBRA), individual (ZEBRA) together with EER on the *VCTK-dev (different)* dataset.

Dataset	WER%		
	Orig	B2	Proposed
LibriSpeech-dev	3.82	4.32	5.88
VCTK-dev	10.79	11.76	15.05
Average Dev Set	7.31	8.04	10.465
LibriSpeech-test	4.15	4.58	5.92
VCTK-test	12.82	13.48	17.88
Average Test Set	8.49	9.03	11.90

Table 5. Primary objective utility evaluation results of metric WER% for ASR_{eval}^{anon} , compared with the original data and the second baseline.

Dataset	N_{sub}	N_{del}	N_{ins}	N_{ref}	WER%
LibriSpeech-dev	1841	227	273	39783	5.88
VCTK-dev	9956	1523	1562	86627	15.05
LibriSpeech-test	1614	214	245	35042	5.92
VCTK-test	11744	2066	1683	86642	17.88

Table 6. Primary objective utility evaluation results of metric WER% for ASR_{eval}^{anon} , with detailed information of the N_{sub} , N_{del} , N_{ins} and N_{ref} .

Dataset	F/M	Weight	p^{F_0}		G_{VD}	
			B2	Proposed	B2	Proposed
LibriSpeech-dev	F	0.25	0.64	0.29	-1.94	-1.36
	M	0.25	0.53	0.21	-1.65	-1.33
VCTK-dev (different)	F	0.20	0.70	0.36	-1.32	-1.38
	M	0.20	0.59	0.28	-2.18	-1.46
VCTK-dev (common)	F	0.05	0.64	0.29	-1.14	-0.99
	M	0.05	0.54	0.27	-1.32	-1.14
Weighted Average Dev Set			0.61	0.28	-1.72	-1.34
LibriSpeech-test	F	0.25	0.61	0.25	-1.71	-1.37
	M	0.25	0.54	0.22	-1.74	-1.30
VCTK-test (different)	F	0.20	0.68	0.28	-1.56	-3.64
	M	0.20	0.66	0.25	-1.56	-1.40
VCTK-test (common)	F	0.05	0.65	0.28	-1.59	-3.16
	M	0.05	0.61	0.26	-1.36	-1.10
Weighted Average Test Set			0.62	0.25	-1.63	-1.89

Table 7. Secondary objective utility evaluation results of metric pitch correlation p^{F_0} and gain of voice distinctiveness G_{VD} , compared with the second baseline.

iv. Secondary Utility Evaluation Results

The results of the secondary utility evaluation, expressed in terms of the scores of p^{F_0} and G_{VD} , are shown in Table 7.

The proposed system failed to achieve the desired pitch correlation p^{F_0} of 0.3 in almost all datasets. This suggests that the anonymised speech has significantly altered the speaker's intonation when compared to the original speech.

Additionally, the suggested system achieved positive results for voice distinctiveness G_{VD} , surpassing all B1 systems [33]. It also outperformed the B2 system on most of the datasets.

iv. ZEBRA Results

The ZEBRA framework was proposed by Nautsch et al. [23] and aims to assess the average level of privacy preservation provided by a specific safeguard across a population, as well as the worst case privacy exposure for an individual.

Figure 7. shows the outcomes of the ZEBRA assessment, with the results of D_{ECE} for the original speech (without protection) omitted. *Perfect privacy* or *zero evidence* (black line) was initially conceptualized as *perfect secrecy* in Shannon's work [28]. The results of the ZEBRA evaluation are presented in the form of

$$(D_{ECE}, \log_{10}(l), \text{tag}) \quad (6)$$

where D_{ECE} is the empirical cross entropy, $\log_{10}(l)$ is the worst-case privacy disclosure, and tag is the categorical tags for disclosure. For further details of the computation process and the meaning of tag classifications, please refer to the ZEBRA paper [23].

The contrast between the ZEBRA outcomes of the B2 system and the proposed system indicates that the performance of the proposed system on the *vctk-o-a-f* and *vctk_common-o-a-f* datasets approaches *perfect privacy*. Furthermore, it outperforms the B2 system and shows a closer proximity to the *perfect privacy* line.

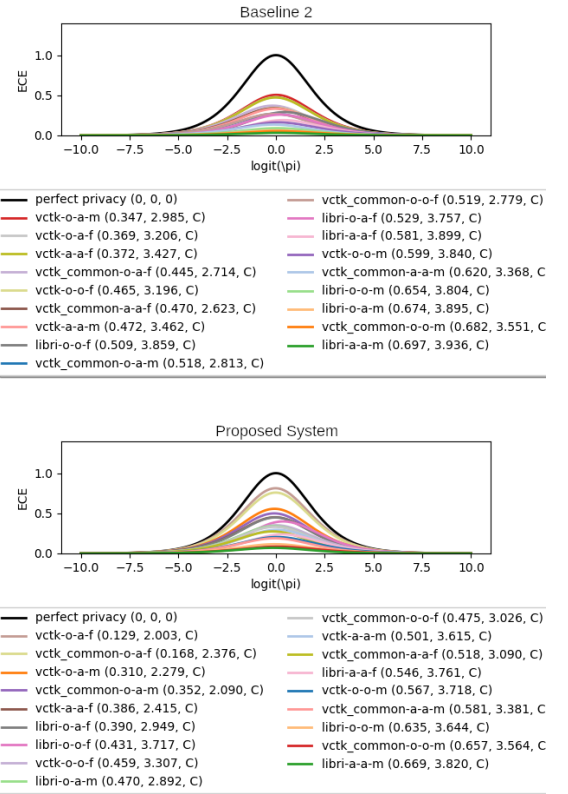


Figure 7. Comparison of ZEBRA results for B2 system and proposed system. The black profile represents *perfect privacy*.

V. CONCLUSIONS

This paper presents the proposed speaker anonymisation system. The primary privacy evaluation results show that the timbre transfer model combined with the McAdams based formant method can reduce the performance of the ASV system by increasing the EER. This result implies an improvement in the privacy of the system. However, it is clear from the results

of the primary utility evaluation that this system sacrifices some of its retention in the WER. The subsequent secondary utility evaluation shows that the proposed system performs well in terms of voice distinctiveness, outperforming all baseline systems. In addition, the primary objective utility evaluation shows potential for further improvement.

VI. FUTURE WORK

Future work includes experimenting with three gender selection strategies, and testing the performance of the system under the three conditions.

Additionally, investigating the impact of gender on speech privacy systems would be of interest, given that previous research indicates that females may possess greater resistance to ASV systems. Therefore, exploring the specific attributes of female speech that contribute to this phenomenon would provide a novel avenue of research. This also suggests whether improved outcomes would be achieved by a biased randomization strategy in a cross-gender timbre transfer.

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